

# Étude du Cancer du Poumon

## Classification des Nodules Pulmonaires par Imagerie CT Scanner

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Pipeline LIDC-IDRI

1139 nodules | 150 patients | 15 features radiomiques

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Introduction à la Data Science

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# 1. Introduction

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Lung cancer is one of the most critical diseases to diagnose early. The aim of this project is to use algorithms to design a decision support system capable of analyzing two types of data sources : behavioral data, which are risk factors, our parameters; and medical imaging (radiological images).

We will use databases not obtained from real patients, due to data confidentiality concerns, but which are sufficient for our student approach to algorithm development.

In the first part, we will use Logistic Regression on a dataset of 309 patients to establish an initial probability based on 15 clinical variables. This step, complemented by a Principal Component Analysis (PCA), allows us to interpret which factors (smoking, age, anxiety) have the greatest impact on the disease. Still using clinical data, we will implement a neural network, which potentially offers greater accuracy.

In the second part, we will use convolutional neural networks (CNNs) to directly process medical images. This phase allows us to detect physical abnormalities that the 15 clinical data points cannot reveal.

# 2. Proposed solutions : the different algorithms used and how they have been generated

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## 2.1 PCA - Principal Component Analysis

The goal of the PCA algorithm is to transform our 15 parameters, which are too complex to visualize, into 2 or 3 main axes. First, we centered and reduced the data so that age wouldn't overwhelm the binary symptoms (0 or 1) in the calculations. Then, we used Singular Value Decomposition (SVD) to find the axes. We eliminated minor variations, those that were too small, to obtain an optimal reduced representation of the dataset.

Similarly, in the second part of our study, PCA served almost the exact same optimization purpose. Just as it helped us reduce the clinical parameters in the first phase, we applied it to compress the highly complex feature space extracted from the lung nodule images. By projecting these high-dimensional features into a lower-dimensional space, we effectively removed noise and redundant information. This ensured that our models could process the most critical patterns efficiently, achieving the same dimensionality reduction and optimization goals across both parts of the project.

## 2.2 LR - Logistic Regression

For Logistic Regression (a single-neuron neural network), we extract the 15 symptoms (variables  $X$ ) and the actual diagnosis ( $y$ ). To allow the model to "shift" its decision boundary, we added a column of 1s to the matrix  $X$ . Furthermore, since we want a probability between 0 and 1, we use the Sigmoid function. Next, we define a cost function  $J(\theta)$  that measures the model's error. While the standard logistic regression algorithm seeks to minimize error on the current data, regularized regression adds the term  $\frac{\lambda}{2m} \sum \theta^2$ .

This prevents the parameters from becoming too large, thereby avoiding overfitting. We used the TNC (Truncated Newton Conjugate) optimization algorithm to automatically find the best parameters  $\theta$ .

Similarly, in the second part of our study, we applied this exact same regularized Logistic Regression approach to serve as our initial baseline model. However, instead of using the 15 clinical symptoms, the input variables ( $X$ ) were the numerical features extracted directly from the lung nodule images (such as radius, texture, and spiculation). Using the same underlying mathematical principles—probabilities optimized through a regularized cost function to prevent overfitting—the algorithm attempted to linearly separate and classify the nodules into three distinct categories : Malignant Cancer, Nodular Irritation, and Benign Cyst. This provided us with a solid, interpretable performance benchmark before transitioning to more complex, non-linear architectures like the MLP and CNN.

## 2.3 MLP - Multi-Layer Perceptron

For the Multi-Layer Perceptron (MLP) neural network algorithm, instead of relying on a single linear calculation (as in Logistic Regression), the architecture consists of multiple interconnected layers of neurons. The input layer receives the extracted numerical features (for example, nodule radius, texture, or spiculation). These inputs are then passed through one or more hidden layers. In these hidden layers, each neuron computes a weighted sum of its inputs, adds a bias term, and applies a non-linear activation function (such as ReLU).

This non-linear combination is crucial, as it allows the network to learn highly complex and abstract patterns from the data that a simple linear model cannot capture. Finally, the processed information reaches the output layer. Since our goal is multi-class classification, this final layer applies a Softmax activation function to output the final probability distribution across our three distinct categories : Malignant Cancer, Nodular Irritation, and Benign Cyst.

## 2.4 CNN - Convolutional Neural Networks

The Convolutional Neural Network (CNN) is used to directly analyze CT scan images of lung nodules, instead of relying on manually defined features. This allows the model to automatically learn relevant visual patterns such as shapes, textures, and irregularities.

To simplify the input, each CT scan is reduced to a small cropped image centered on the nodule. This region is then converted into three different views using Hounsfield windowing (lung, mediastinum, and bone), which enhances different anatomical structures. These three views are combined as channels, similarly to a color image.

The CNN processes these images through several convolutional layers, which detect local patterns, followed by pooling and normalization steps that reduce dimensionality and improve stability. As the information flows through the network, the extracted features become increasingly abstract and informative.

Finally, the network outputs a probability indicating the likelihood that the nodule is malignant. This probability is then used to classify the nodule into different diagnostic categories.

### 3. Results obtained by the learning

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#### 3.1 PCA - Principal Component Analysis

Principal Component Analysis projects the data onto a lower-dimensional space by maximizing variance. It is based on the covariance matrix :

$$\Sigma = \frac{1}{N} X^T X$$

and consists in solving the eigenvalue problem :

$$\Sigma w = \lambda w$$

The projection of the data is then given by :

$$Z = XW$$

##### Interpretation of our results :

The PCA grouped the symptoms into "families" : the Z1 axis is linked to behavioral and emotional symptoms (Anxiety, peer pressure, smoking), while the Z2 axis is related to physical and respiratory symptoms (Coughing, shortness of breath, wheezing).

Furthermore, by reducing the dimensions from 15 to 2, we can visualize the data on a 2D plot and observe a clear separation between healthy patients and sick patients.

Similarly, for the second part of our study, the PCA grouped the extracted image features into logical "families" : the Z1 axis is linked to morphological characteristics (Diameter, volume, and compactness), while the Z2 axis is related to textural and margin features (Spiculation, calcification, and intensity).

Furthermore, by reducing the high-dimensional image features down to 2 dimensions, we were able to visualize the data on a 2D plot and observe a clear separation between our three categories : Malignant Cancers, Nodular Irritations, and Benign Cysts.

#### 3.2 LR - Logistic Regression

Logistic Regression models the probability of a binary outcome using the sigmoid function :

$$P(y = 1|x) = \frac{1}{1 + e^{-\theta^T x}}$$

The parameters  $\theta$  are learned by minimizing a cost function (typically cross-entropy), allowing the model to assign weights to each feature.

##### Interpretation of our results :

Logistic (and Regularized) Regression : This algorithm achieved an accuracy of 94.5

For the regularized logistic regression model, testing various values of  $\lambda$  (lambda) shows that our model is robust. Regularization ensures that the  $\theta$  weights assigned to symptoms (such as smoking or anxiety) are not disproportionate, which limits the risk of overfitting on this small sample of 309 patients.

The algorithm "learned" which symptoms are the most severe. By examining the  $\theta$  values, we can see that certain factors increase the probability of cancer much faster than others.

Similarly, for the second part of our study, the Regularized Logistic Regression served as our baseline model and achieved an accuracy of 74.0% on a blind test set of 50 independent patients. This means that the algorithm correctly classified roughly 3 out of 4 nodules into their true categories (Malignant Cancer, Nodular Irritation, or Benign Cyst). While this accuracy is lower than in Part 1, it is entirely expected : attempting to separate complex, non-linear image features with a simple linear boundary is inherently more challenging than analyzing binary clinical symptoms.

Just as in the first phase, testing various values of  $\lambda$  (lambda) demonstrated the model's robustness. Regularization ensured that the  $\theta$  weights assigned to the extracted image features (such as nodule radius, texture, or spiculation) did not become disproportionate. This strictly limited the risk of overfitting on the training data, allowing us to establish a reliable, mathematically sound baseline before advancing to more complex architectures like the MLP and CNN.

### 3.3 MLP - Multi-Layer Perceptron

A Multi-Layer Perceptron (MLP) models complex non-linear relationships by composing multiple layers of transformations :

$$f(x) = \sigma(W_2 \sigma(W_1 x + b_1) + b_2)$$

where  $\sigma$  is a non-linear activation function (e.g., ReLU), and  $W$ ,  $b$  are the learnable parameters.

#### Interpretation of our results :

Neural Network Algorithm (MLP) : We observe that the MLP algorithm has a higher accuracy score than logistic regression. Indeed, where the regression drew a straight line (linear boundary), the neural network is capable of drawing complex curves to better encompass patient groups. This demonstrates increased flexibility, although it requires constant monitoring for overfitting.

Similarly, in the second part of our study, the implementation of this MLP architecture yielded a significant performance boost, achieving an accuracy of 80.0% on our blind test set (surpassing the 74.0% of our baseline). By utilizing hidden layers (configured with 256 and 128 neurons) and ReLU activations, the network successfully captured the non-linear relationships within the extracted image features. For instance, the MLP could mathematically understand complex combinations—such as a nodule having a small radius but high spiculation—to effectively separate the data into our three categories (Malignant Cancer, Nodular Irritation, and Benign Cyst) far better than the rigid linear boundaries of the previous model.

### 3.4 CNN - Convolutional Neural Networks

#### Interpretation of our results :

The CNN proved to be the best model in our study, achieving a top accuracy of 84.0% on the blind test. Unlike previous algorithms, it directly analyzes the raw 48x48 pixel images (using three clinical windows : lung, mediastinum, and bone) instead of relying on pre-calculated numbers. By automatically learning visual patterns from the scans, it significantly outperformed both Logistic Regression and MLP.

Finally, we applied the **Grad-CAM** algorithm to make the CNN interpretable. This tool generates visual heatmaps showing exactly which parts of the image influenced the model's prediction, ensuring our diagnosis is not only highly accurate but also clinically transparent.

## 4. Conclusion

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This project successfully established a decision-support pipeline for early lung cancer diagnosis, leveraging the complementarity between clinical data and medical imaging. We began by applying Principal Component Analysis (PCA) to simplify biological complexity, effectively reducing 15 clinical symptoms into interpretable behavioral and physical axes. This dimensionality reduction provided a clear visual separation between healthy and sick patients right from the exploratory phase.

Transitioning to predictive modeling, our progressive approach validated the need for increasingly complex architectures. The Regularized Logistic Regression served as a highly transparent baseline, achieving 94.5% accuracy on clinical data (Part I) and 74.0% on extracted image features (Part II) while successfully limiting overfitting. To capture more complex, non-linear interactions, we implemented a Multi-Layer Perceptron (MLP), which improved performance to approximately 82%. Ultimately, the Convolutional Neural Network (CNN) proved to be the core of our pipeline. By directly processing  $48 \times 48$  CT crops across three Hounsfield windows (Lung, Mediastinum, Bone), the CNN mimicked a radiologist's visual perspective, extracting intricate textural patterns to reach the highest accuracy of 84.0%.

This progression clearly demonstrates the superiority of deep learning for raw medical image processing :

$$\text{Logistic Regression (74\%)} < \text{MLP (\sim 82\%)} < \text{CNN (84\%)}$$

**Limitations :** Despite these strong performance scores, caution is required due to the dataset's high class imbalance (87% positive cases in the initial phase). A model might achieve high accuracy simply by over-predicting the majority class. This highlights the absolute necessity of relying on complementary metrics, such as precision, recall, and the F1-score, to ensure a truly robust clinical validation in future work.

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